ALTEGRAD Data Challenge : Molecular Graph Captioning Report

Adam Rozzio Emett Haddad

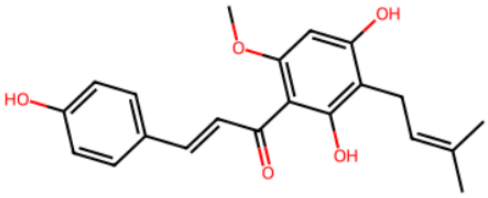
January 25, 2026

Abstract

In the field of cheminformatics, accurately translating complex 2D structures of molecules into their textual descriptions is a challenging yet essential task for scientific literature, augmenting chemical database entries, and assisting in AI-driven drug discovery pipelines. This project aims to address this challenge by testing different models for this multi-modal graph-to-text generation task. We explored several distinct approaches to address this problem, including different model architectures, SMILES representations, graph neural network (GNN)–based encodings of molecular graphs, an encoder–decoder framework using T5ForConditionalGeneration, and RoBERTa-base to encode textual representations.

1 Introduction

In this report, we present our work on the Molecular Graph Captioning Project (ALTEGRAD private competition, 2025), where we are seeking to generate a description based on the molecule's graph.



Description: The molecule is a member of the class of chalcones that is trans-chalcone substituted by hydroxy groups at positions ...

Figure 1: A molecule and it's description (id: 9, training dataset)

This difficulty of such a task comes from the the differences in the representation of textual and graph data. To bridge this gap we will consider **retrieval and generation techniques**. In the first case, for a given encoded molecule graph one aims a finding the closest encoded molecule caption which will be then attributed to the graph and used for evaluation. On the other hand the the generative task does not encode the caption but instead outputs a molecule caption from an input molecule graph. This report explores different approaches for each of these strategies based on techniques discussed both in class (word embedding, fine-tuning an LLM with LoRa) and in the literature (molecule graph embedding and multi-modal molecule representation [28, 32, 17]). Regarding the structure of the report we will first mention the existing works on this topic on the literature before specifying the problem definition. Then, we will detail the tested methods and architectures and finally present the experimental results.

2 State of the art

In early 2026, there have been many attempts at bridging the gap between textual and graph representation of molecules. The first methods to be developed were the retrieval approaches which we recall are based on a common embedding of molecules graph and captions.

Molecules Graph Embedding

The pioner work on molecules graph representation in 1988 is [28].It introduces 1D sequences representation called SMILES, such representation is not unique however it is common to consider the canonical SMILES representation. More recently, the efforts have been directed towards Graph Neural Networks architectures [24] and some examples of this class are presented in 4.1.2. Nowadays, the tendency is to integrate other modalities to the encoding of a GNN to accumulate different information to enrich the embedding, leading to higher performances [17].

Molecules Caption Embedding

The word embedding task has first raised attention with Word2Vec [18] and GloVe [20] before being dominated by contextual dependant embedding methods like BERT [6] and other versions fine-tuned on Scientific text like SciBERT [1] or on

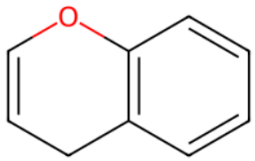


Figure 2: A molecule and it's SMILES representation: C1=COc2ccccc2C1 (id:11, training dataset). We used rdkit to generate SMILES representations from graphs.

MoleculeNet [29] like ChemBERTa [5].

2.1 Retrieval methods

The retrieval methods methods were the first to emerge in 2021 with Text2Mol [8], the first work to apply the ideas of contrastive learning used in CLIP [21] to molecules. In particular they considered SciBERT [1] to encode the molecules captions and a Graph Convolutional Network (GCN) [13] to embed the graph. Similarly [26] introduced a Molecular Multimodal (MoMu) fundation model considering a Graph Isomorphism Network (GIN) [30] to encode the molecules and BERT as the text encoder. Recently in 2025 the frontier has shifted towards fine-grained alignment with models like Atomas [34], which align substructures of the graph with specific phrases in the text instead of using a single global vector, it also exploits the SMILES information.

2.2 Generative methods

The first major milestone was MolT5 [9], which treated molecule-to-text as a translation task. By fine-tuning a T5 architecture on SMILES-caption pairs, it demonstrated that language models could internalize chemical grammar. However, relying solely on 1D SMILES often fails to capture complex topological nuances. To address this, MolCA [17] introduced a cross-modal projector (Q-Former) to link a GNN with a frozen LLM (Galactica) effectively injecting 2D structural information into the generative process.

In parallel, GitMol [16] expanded this by proposing a multi-modal approach that fuses SMILES, graphs, and even 2D images into a single transformer-based space. More recently, GraphT5 [12] (2025) has pushed the boundaries of this integration by using a *Cross-Token Attention* mechanism. Unlike previous methods that simply concatenated embeddings, GraphT5 allows for a fine-grained alignment between the SMILES tokens and the graph nodes, leading to more accurate and chemically consistent captions.

A recent model XMolCap (2025) [27] has established that combining SMILES and molecule images yields the highest performance on ChEBI-20 benchmarks. These architectures typically leverage a vision transformer (ViT) [7] alongside a chemical LLM, using the image to enrich the model's understanding of the molecule before refining the description with precise SMILES based details.

3 Problem

3.1 Dataset

The dataset contains approximately 33000 molecule-text pairs, split as follows: training and validation set are composed of about 32000 molecules, provided with both their full graph structure and their corresponding ground-truth caption. The rest is the test set, composed of 1000 molecules provided with their graph structure only.

3.2 Evaluation

The evaluation is a composite score derived from a precision-oriented metric, BLEU-4 [19] F1 Score, that measures the overlap of n-grams (word sequences) between the generated and reference texts and a modern, semantic-focused metric, and BERTScore [33] with RoBERTa-base , that computes the cosine similarity between the embeddings of tokens in the generated and reference texts, using a pre-trained BERT-style model.

4 Method

In this section, we detail the specific architectures and methodologies employed to tackle the graph-to-text generation and retrieval task. For the generation we adopted an Encoder-Decoder framework where a Graph Neural Network (GNN) encodes the molecular structure, and a Language Model (LLM) generates the corresponding textual description. On the other hand the retrueval task was achieved by encoding the graphs with a GNN and encoding the caption with a Transformer

4.1 Architecture

4.1.1 Text Generation and Text Embedding Models

To handle the generation of chemical descriptions, we leveraged several pre-trained language models. The choice of the decoder is crucial as it must capture the specific syntax and terminology of chemical literature.

- **T5 for Conditional Generation**: We primarily utilized the Text-to-Text Transfer

Transformer (T5) [22]. T5 is an encoder-decoder Transformer pre-trained on a multi-task mixture of unsupervised and supervised tasks, where every task is converted into a text-to-text format. This architecture achieved state-of-the-art results in transfer learning for NLP in recent years. It uses SentencePiece [14] to encode text as WordPiece tokens, allowing it to handle rare chemical tokens effectively.

- **BERT-based Models**: We also explored encoder-only architectures for the retrieval task. We tested SciBERT [1], which is trained on a large corpus of scientific text, DistilBERT [23] for a lighter and faster alternative, and ChemBERT [5], a RoBERTa model pre-trained specifically on SMILES strings from the ZINC dataset.

4.1.2 Convolutional Layers

The Molecular Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ encoding part of the retrieval task relies on Graph Neural Networks. We experimented with several Message Passing mechanisms available in the PyTorch Geometric library to capture local and global structural information. In the following h_θ is an MLP with with θ as parameters and $\mathcal{V}(i)$ the indexes of the neighbours of the node indexed by i.

- **TransformerConv** [25]: It applies the attention mechanisms of the Transformer architecture to graphs and weights the importance of neighbors : $\mathbf{x}'_i = \mathbf{W}_1\mathbf{x}_i + \sum_{j\in\mathcal{V}(i)} \alpha_{i,j}\mathbf{W}_2\mathbf{x}_j$. Where $\alpha_{i,j}$ are the attention coefficients computed via softmax.

- **GeneralConv**: Based on [31], GeneralConv is a general GNN layer that supports various message passing aggregators (mean, max, add) and can approximate other GNN layers.

- **GINEConv**: We considered the Graph Isomorphism Network with Edge features (GINE) [11], designed to incorporate bond information. It is known for its discriminative power regarding graph structures. The update rule is defined as

$$\mathbf{x}_i^{'} = h_\theta\big(\mathbf{x}_i + \sum_{j\in\mathcal{V}(i)} \mathrm{ReLU}(\mathbf{x}_j + \mathbf{e}_{j,i})\big).$$

4.1.3 Aggregation Operators

After processing the graph through convolutional layers, we must aggregate the individual node embeddings into a single fixed-size vector $\mathbf{x}_\mathcal{G}$ representing the entire molecule. We note $\mathcal{V}$ the indexes of the nodes of the graph. We explored the following pooling strategies:

- **MaxAggregation**: By performing element-wise maximum pooling, this operator identifies the most representative features regardless of the graph size : $\mathbf{x}_\mathcal{G} = \max_{n\in\mathcal{V}} \mathbf{x}_n$

- **MeanAggregation**: It captures the average distribution of atomic properties across the molecular graph : $\mathbf{x}_\mathcal{G} = \frac{1}{|\mathcal{V}|}\sum_{n\in\mathcal{V}} \mathbf{x}_n$.

- **DeepSetsAggregation** [3]: Based on the framework of DeepSets, this operator ensures permutation invariance by summing transformed node features before passing them through a MLP : $\mathbf{x}_\mathcal{G} = h^{(2)}_{\theta_2}\left(\sum_{n\in\mathcal{V}} h^{(1)}_{\theta_1}(\mathbf{x}_n)\right)$

- **AttentionalAggregation** [15]: It applies a soft attention layer to learn importance scores for each node, allowing the model to focus on substructures relevant to the captioning task : $\mathbf{x}_\mathcal{G} = \sum_{n\in\mathcal{V}} \mathrm{softmax}(h_\theta(\mathbf{x}_n)) \odot \mathbf{x}_n$

4.2 Contrastive Learning

To enable the retrieval task, we aim to learn a joint embedding space where the representation of a molecular graph is close to the representation of its corresponding textual description. We employ contrastive learning strategies to align these two modalities, as reviewed in [10].

Similarity Metric. We consider the cosine similarity to measure the proximity between a graph embedding vector u and a text embedding vector v:

$$\mathrm{sim}(u, v) = \frac{u^\top v}{\|u\|\,\|v\|}$$

InfoNCE Loss. For the training objective, we adopt the InfoNCE loss, popularized by CLIP [21]. Given a batch of N molecule-text pairs, the goal is to maximize the similarity of the correct pairs while minimizing the similarity of the incorrect pairings. For a specific graph embedding g_i and text embedding t_i, the graph-to-text loss is defined as:

$$\mathcal{L}_{G\to T} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp\left(\mathrm{sim}(g_i, t_i)/\tau\right)}{\sum_{j=1}^{N} \exp\left(\mathrm{sim}(g_i, t_j)/\tau\right)}$$

where τ is a temperature hyper parameter. The symmetric text-to-graph loss $\mathcal{L}_{T\to G}$ is calculated similarly. The final objective function is the average of both directions $\mathcal{L}_{\text{contr}} = \frac{1}{2}(\mathcal{L}_{G\to T} + \mathcal{L}_{T\to G})$

Triplet Loss. We also considered Triplet Loss, which operates on triplets (anchor a, positive p, negative n) and enforces the margin constraint $\mathcal{L}_{\text{t}} = \max(0, d(a, p) - d(a, n) + \alpha)$. In particular it helped the InfoNCE loss to deal with the local hard cases.

Soft Triplet Loss. To remove an hyper parameter and let the training phase continue we considered the Soft Triplet Loss : $\mathcal{L}_{\text{st}} = \log(1 + \exp(d(a,p) - d(a,n))$.

5 Experiments

We noticed that using a small language model to generate text ultimately yielded the same performance/generation capabilities as a model that retrieves the closest text. We therefore submitted our best T5 model as well as our best retrieval model.

5.1 Retrieval model

We display the structure of the retrieval model in Figure 3. To get the best out of this model we lead experiments in Appendix to identify the right structure : TransformerConv as the convolutional layer in Figure 9), AttentionalAggregation to aggregate the node features in Table 2, 256 for the hidden dimension in Table 3 and 5 for the number of layers in Table 4. Additionnaly we add a 1 layer MLP on the top of BERT to embed the captions to exploit the full strength of the contrastive learning strategy. This setting with $\mathcal{L} = \mathcal{L}_{\rfloor} + 0.1\mathcal{L}_{\int}\sqcup$lead to a score of 0.645 on Kaggle.

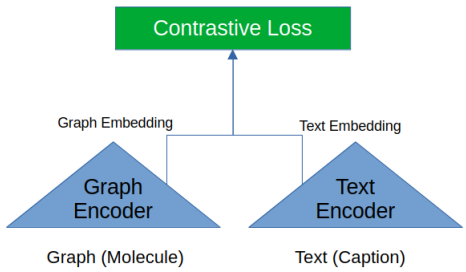


Figure 3: Retrieval model

5.2 Generative models using T5

We tested fine tuning with LoRA and freezing parts of T5, but ultimately the most effective approach seemed to be retraining it completely from the pretrain t5-small model. We noticed that T5 tends to **overfit** the captions in the training dataset, with validation loss increasing as the Kaggle evaluation improves with model training, and the model learning exactly the captions in the dataset, even learning the web links of certain articles related to the molecules, despite the use of dropout. We used **t5-small** as a text generator, along with its default training loss and Adam as an optimizer. More details on layer sizes/architectures can be found in the various code directories, however we generally used: a hidden dimension of around 128, 3 layers of the GNN and attentional aggregation with GINEConv convolution layers.

5.2.1 SMILES + T5

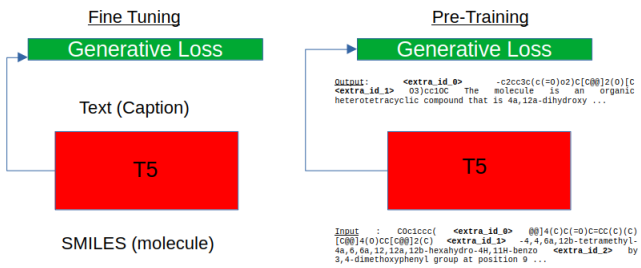


Figure 4: SMILES to LLM model

We achieve a score of 0.659 using this model with the pretrained t5-small model, that we retrain to replace corrupted spans during 8 hours and fine-tune (with LoRA) it during 12 hours with 24G of VRAM.

5.2.2 GNN + T5

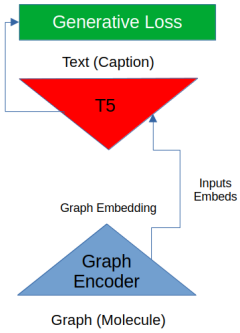


Figure 5: Generative model

With this simple model, we achieved a score of 0.64 on Kaggle, after a training of 12 hours with 24G of VRAM. We present in Table 1 the results of a small experiment.

Description	Generation
The molecule is a hydroxyisoflavanone that is isoflavanone substituted by hydroxy groups ...	The molecule is a dihydroxyflavanone that is flavanone substituted by hydroxy groups ...
The molecule is a fluorobenzoic acid carrying a fluoro substituent at position 4 ...	The molecule is a monocarboxylic acid anion resulting from the removal of a proton ...

Table 1: Good and bad examples with this model

5.2.3 SMILES + GNN + T5

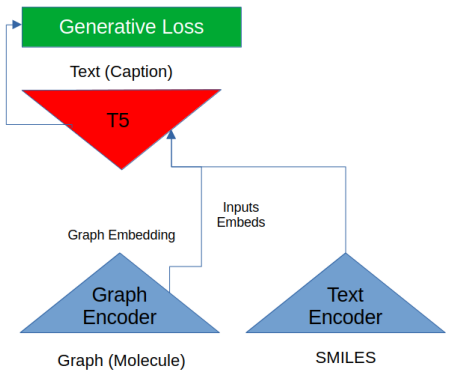
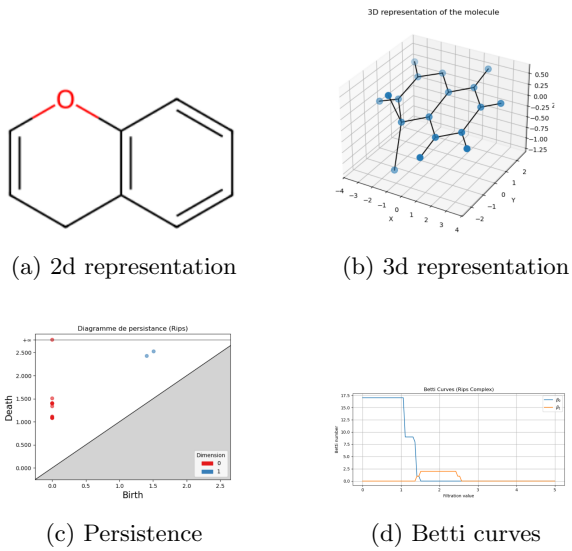


Figure 6: SMILES and graph to LLM model

We achieved a score of 0.604 with this method. The principle behind this method, is to enrich the data obtained from the GNN on the molecule graph by adding the embedding of the SMILES representation of the molecules. Although this method is very effective, we have not been able to exploit its full potential in our experiments.

6 Opening: Unfinished ideas

6.1 Betti curves



(a) 2d representation (b) 3d representation

(c) Persistence (d) Betti curves

Figure 7: Betti curves for a molecule (train dataset, id:11), a) SMILES representation, b) 3d representation of the molecule after physical optimisation of chemical bond, c) Persitence Diagram of the molecule, d) associated betti curves

We could use this Gudhi software to compute **Betti Curves** [4], a featurisation of the molecule's persistence diagram which has already demonstrated its effectiveness in numerous areas of application. For a persistence diagram D (multiset $D = \{(b_i, d_i), i \in I\}$), the formula is $B(D) = \sum_{i \in I} \mathbf{1}_{[b_i, d_i]}$. This would enable long cycles and other topological features of molecules to be captured efficiently, and often proves to be quite complementary to GNN. We could have used spatial Rips Complex to generate the persistence diagram.

6.2 Data pre-processing

When manipulating the dataset, it can be seen that there is still some extraneous data in the descriptions, such as HTML links or references to articles related to the molecule. We have implemented a small algorithm based on t5-large [22] that **cleans up descriptions** by instructing it to : "Clean the following molecule description by removing web links and non-structural information such as discovery, authors, dates, institutions or publications.". However, it turns out that performing such preprocessing would require an excessive amount of time—exceeding ten hours—making it impractical.

6.3 SMILES augmentation

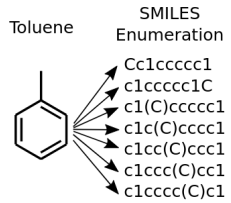


Figure 8: SMILES enumerations, figure from [2]

Simple strategy from [2], which consists of using **enumerations of all SMILES representations** as input to the textual decoder rather than just the canonical representation.

7 Conclusion

In this project, we investigated both retrieval-based and generative approaches for molecular graph captioning, demonstrating that encodings using graph neural networks or textual representations can effectively bridge the gap between molecular structure and natural language. Our experiments show that, despite their simplicity, retrieval models remain competitive with generative architectures with our training dataset, as T5-based generative models achieve stronger semantic alignment at the cost of overfitting and higher computational requirements. We also discussed promising directions, including topological descriptors, data cleaning, and richer SMILES-based augmentations to improve generalization.

References

[1] Iz Beltagy, Kyle Lo, and Arman Cohan. "SciBERT: A pretrained language model for scientific text". In: *arXiv preprint arXiv:1903.10676* (2019).

[2] Esben Jannik Bjerrum. "SMILES enumeration as data augmentation for neural network modeling of molecules". In: *arXiv preprint arXiv:1703.07076* (2017).

[3] David Buterez et al. "Graph neural networks with adaptive readouts". In: *Advances in Neural Information Processing Systems* 35 (2022), pp. 19746–19758.

[4] Frédéric Chazal and Bertrand Michel. "An introduction to topological data analysis: fundamental and practical aspects for data scientists". In: *Frontiers in artificial intelligence* 4 (2021), p. 667963.

[5] Seyone Chithrananda, Gabriel Grand, and Bharath Ramsundar. "ChemBERTa: large-scale self-supervised pretraining for molecular property prediction". In: *arXiv preprint arXiv:2010.09885* (2020).

[6] Jacob Devlin et al. "Bert: Pre-training of deep bidirectional transformers for language understanding". In: *Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers)*. 2019, pp. 4171–4186.

[7] Alexey Dosovitskiy. "An image is worth 16x16 words: Transformers for image recognition at scale". In: *arXiv preprint arXiv:2010.11929* (2020).

[8] Carl Edwards, ChengXiang Zhai, and Heng Ji. "Text2mol: Cross-modal molecule retrieval with natural language queries". In: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*. 2021, pp. 595–607.

[9] Carl Edwards et al. "Translation between molecules and natural language". In: *arXiv preprint arXiv:2204.11817* (2022).

[10] Haigen Hu et al. "A comprehensive survey on contrastive learning". In: *Neurocomputing* 610 (2024), p. 128645.

[11] Weihua Hu et al. "Strategies for pre-training graph neural networks". In: *arXiv preprint arXiv:1905.12265* (2019).

[12] Sangyeup Kim et al. "GraphT5: Unified Molecular Graph-Language Modeling via Multi-Modal Cross-Token Attention". In: *arXiv preprint arXiv:2503.07655* (2025).

[13] TN Kipf. "Semi-supervised classification with graph convolutional networks". In: *arXiv preprint arXiv:1609.02907* (2016).

[14] Taku Kudo and John Richardson. "SentencePiece: A simple and language independent subword tokenizer and detokenizer for neural text processing". In: *arXiv preprint arXiv:1808.06226* (2018).

[15] Yujia Li et al. "Graph matching networks for learning the similarity of graph structured objects". In: *International conference on machine learning*. PMLR. 2019, pp. 3835–3845.

[16] Pengfei Liu et al. "Git-mol: A multi-modal large language model for molecular science with graph, image, and text". In: *Computers in biology and medicine* 171 (2024), p. 108073.

[17] Shengchao Liu et al. "Multi-modal molecule structure–text model for text-based retrieval and editing". In: *Nature Machine Intelligence* 5.12 (2023), pp. 1447–1457.

[18] Tomas Mikolov et al. "Distributed representations of words and phrases and their compositionality". In: *Advances in neural information processing systems* 26 (2013).

[19] Kishore Papineni et al. "Bleu: a method for automatic evaluation of machine translation". In: *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*. 2002, pp. 311–318.

[20] Jeffrey Pennington, Richard Socher, and Christopher D Manning. "Glove: Global vectors for word representation". In: *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*. 2014, pp. 1532–1543.

[21] Alec Radford et al. "Learning transferable visual models from natural language supervision". In: *International conference on machine learning*. PmLR. 2021, pp. 8748–8763.

[22] Colin Raffel et al. "Exploring the limits of transfer learning with a unified text-to-text transformer". In: *Journal of machine learning research* 21.140 (2020), pp. 1–67.

[23] Victor Sanh et al. "DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter". In: *arXiv preprint arXiv:1910.01108* (2019).

[24] Franco Scarselli et al. "The graph neural network model". In: *IEEE transactions on neural networks* 20.1 (2008), pp. 61–80.

[25] Yunsheng Shi et al. "Masked label prediction: Unified message passing model for semi-supervised classification". In: *arXiv preprint arXiv:2009.03509* (2020).

[26] Bing Su et al. "A molecular multimodal foundation model associating molecule graphs with natural language". In: *arXiv preprint arXiv:2209.05481* (2022).

[27] Duong Thanh Tran et al. "XMolCap: Advancing Molecular Captioning through Multimodal Fusion and Explainable Graph Neural Networks". In: *IEEE Journal of Biomedical and Health Informatics* (2025).

[28] David Weininger. "SMILES, a chemical language and information system. 1. Introduction to methodology and encoding rules". In: *Journal of chemical information and computer sciences* 28.1 (1988), pp. 31–36.

[29] Zhenqin Wu et al. "MoleculeNet: a benchmark for molecular machine learning". In: *Chemical science* 9.2 (2018), pp. 513–530.

[30] Keyulu Xu et al. "How powerful are graph neural networks?" In: *arXiv preprint arXiv:1810.00826* (2018).

[31] Jiaxuan You, Zhitao Ying, and Jure Leskovec. "Design space for graph neural networks". In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 17009–17021.

[32] Zheni Zeng et al. "A deep-learning system bridging molecule structure and biomedical text with comprehension comparable to human professionals". In: *Nature communications* 13.1 (2022), p. 862.

[33] Tianyi Zhang et al. "Bertscore: Evaluating text generation with bert". In: *arXiv preprint arXiv:1904.09675* (2019).

[34] Yikun Zhang et al. "Atomas: Hierarchical adaptive alignment on molecule-text for unified molecule understanding and generation". In: *The Thirteenth International Conference on Learning Representations*. 2025.

8 Appendix

8.1 Numerical Experiments on The Retrieval Task

For this task the experiments were run locally on on an NVIDIA GeForce RTX 4060 Laptop GPU (8GB VRAM). Every training last approximately an hour.

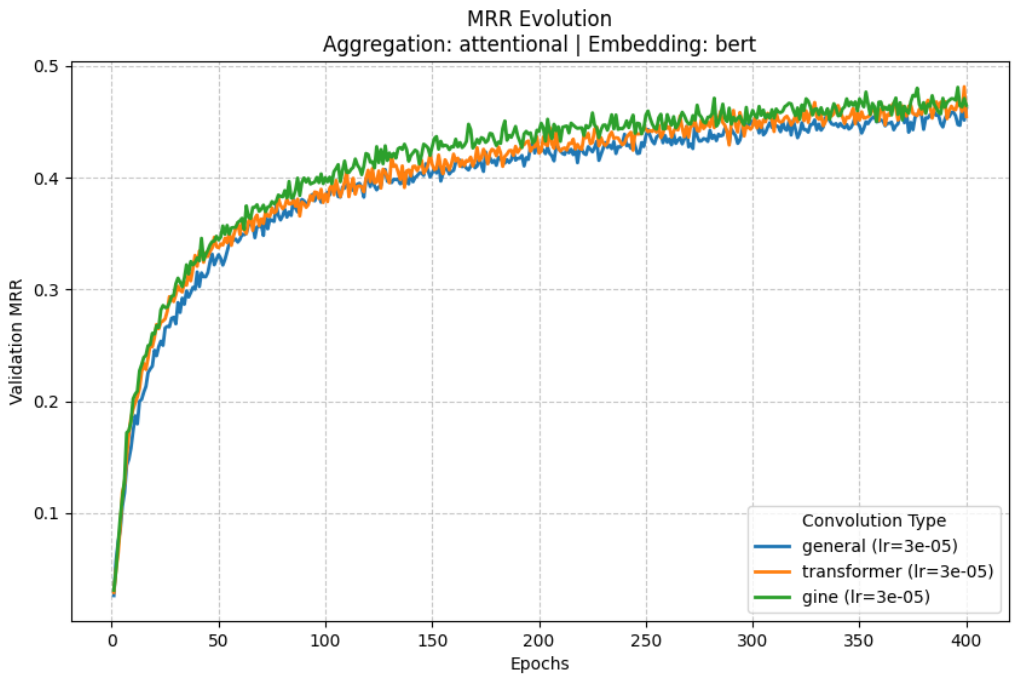


Figure 9: Convolution Layer using the retrieval model.

Aggregation Operator	Loss / MRR	Hit10
DeepSetsAggregation [3]	3.13/0.453	0.661
AttentionalAggregation [15]	**3.13/0.466**	0.662
MeanAggregation	3.14/0.456	0.657
MaxAggregation	3.14/0.450	0.651

Table 2: Test of the impact of different aggregation operators for the GNN on the results, using the GNN + T5 model, with minimal training.

Hidden dimension	Loss / MRR	Evalutation
64	3.34/0.399	0.5308
128	3.21/0.428	**0.5398**
256	**3.08/0.464**	0.5391
512	3.93/0.155	0.5095

Table 3: Test of the impact of the hidden dimension of the GNN on the results, using the GNN + T5 model, with minimal training.

Number of layers	Loss / MRR	Hit10
3	2.30/0.4796	0.679
4	2.27/ 0.4827	**0.688**
5	**2.24/0.4910**	0.679
6	2.23/0.4760	0.684

Table 4: Test of the impact of the number of layers of the GNN on the results, using the GNN + T5 model, with minimal training.

8.2 Numerical Experiments on The Generation Task

For this task the experiments were run locally on on an NVIDIA GeForce RTX 5090 Laptop GPU (24GB VRAM).

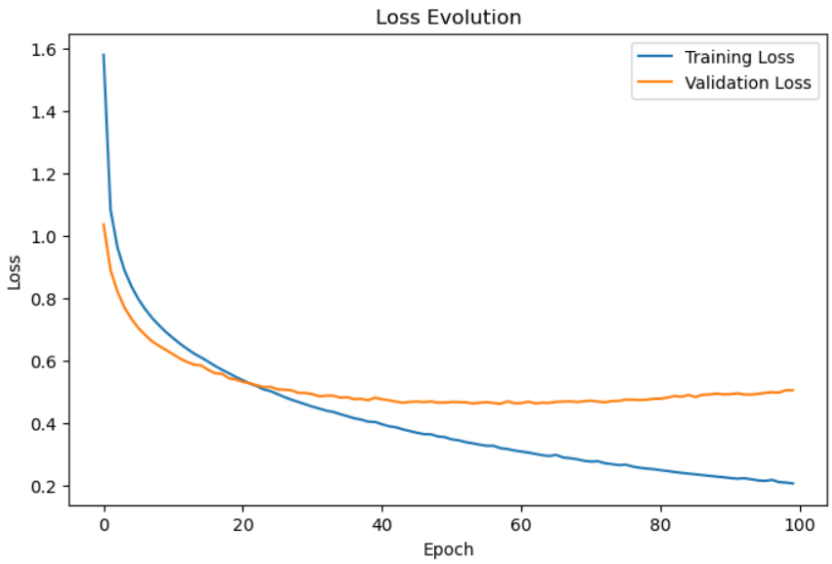


Figure 10: Loss training and validation for the GNN + T5 model

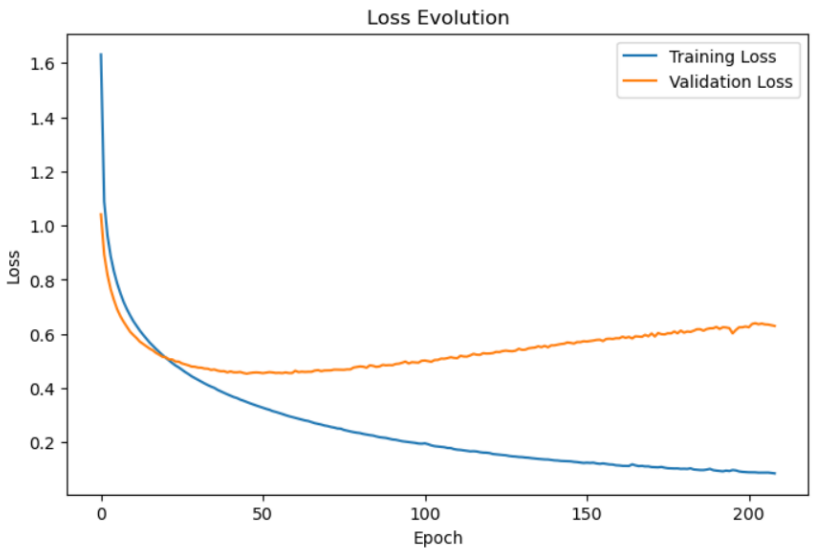


Figure 11: Loss training and validation for the SMILES + GNN + T5 model